

The Helical Arithmetic of the 13_3 Microtubule Lattice is Governed by χ_3

A two-lemma note

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Abstract

We model the canonical 13-protofilament, 3-start microtubule surface lattice as the cylindrical quotient $L = \mathbb{Z}^2 / \langle (-3, 13) \rangle$ and prove that the number of disjoint strands in the helical family of any primitive direction (p, q) is the lattice index $n(p, q) = |13p + 3q|$ (Lemma 1). The two coefficients of this form are exactly the arithmetically distinguished primes of the Eisenstein field $\mathbb{Q}(\sqrt{-3})$: the protofilament number $13 = N(4 + \omega)$ splits, and the start number $3 \sim (1 - \omega)^2$ ramifies (Lemma 2). Consequently the start-number law is a χ_3 -form; the 3-start helix realises the ramified prime $(1 - \omega)$; and its three-fold screw symmetry is the order-3 rotation $\langle \omega \rangle$ of the Eisenstein units, i.e. the “tri-phase” $\{1, \omega, \omega^2\}$. The Fibonacci indices 5, 8 that recur in the microtubule folklore are, by contrast, inert ($\chi_3 = -1$) and carry no distinguished status in the lattice.

1 The cylindrical lattice

Let a microtubule consist of $N = 13$ protofilaments arranged with a $c = 3$ start helix (the 13_3 lattice observed *in vivo*). Unroll the cylinder onto the plane and let the surface lattice be generated by

$$\mathbf{a}_1 = (0, h), \quad \mathbf{a}_2 = (s, \frac{c}{N}h) = (s, \frac{3}{13}h),$$

where $\mathbf{a}_1$ is the longitudinal (protofilament) step of monomer rise h and $\mathbf{a}_2$ is the lateral step to the adjacent protofilament. Index the subunits by $\mathbb{Z}^2$ via $(m, n) \mapsto m\mathbf{a}_1 + n\mathbf{a}_2$.

Closing the cylinder identifies physical points that differ by one full turn. A single turn is 13 lateral steps, which advances axially by 3 monomers; subtracting those 3 longitudinal steps yields a *pure* circumferential translation

$$\mathbf{W} = -c\mathbf{a}_1 + N\mathbf{a}_2 = -3\mathbf{a}_1 + 13\mathbf{a}_2, \quad \text{i.e. } \mathbf{W} = (-3, 13) \in \mathbb{Z}^2.$$

Hence the microtubule surface lattice is the cylindrical quotient

$$L = \mathbb{Z}^2 / \langle \mathbf{W} \rangle, \quad \mathbf{W} = (-c, N) = (-3, 13).$$

Definition 1 (Helical family and start number). For a primitive vector $\mathbf{v} = (p, q) \in \mathbb{Z}^2$ (i.e. $\gcd(p, q) = 1$), the *helical family* $\mathcal{H}_{\mathbf{v}}$ is the partition of L into the orbits of translation by $\mathbf{v}$. Each orbit is one *strand*; the *start number* $n(\mathbf{v})$ is the number of strands, i.e. the number of distinct parallel helices needed to cover all subunits.

2 Lemma 1: the start–number law

Lemma 1 (Start–number law). *For every primitive $\mathbf{v} = (p, q) \in \mathbb{Z}^2$,*

$$n(\mathbf{v}) = |\det[\mathbf{v}, \mathbf{W}]| = |Np + cq| = |13p + 3q|.$$

Proof. Two subunits lie on the same strand of $\mathcal{H}_{\mathbf{v}}$ iff their difference in L is a $\mathbb{Z}$ –multiple of $\mathbf{v}$; that is, iff their representatives in $\mathbb{Z}^2$ differ by an element of the subgroup $\Lambda = \langle \mathbf{v}, \mathbf{W} \rangle \leq \mathbb{Z}^2$. The strands are therefore the cosets of Λ , so

$$n(\mathbf{v}) = [\mathbb{Z}^2 : \Lambda] = |\det[\mathbf{v}, \mathbf{W}]|,$$

because $\mathbf{v}$ and $\mathbf{W}$ are linearly independent ($\mathbf{v}$ has nonzero axial component while $\mathbf{W}$ is purely circumferential). Evaluating the determinant with $\mathbf{v} = (p, q)$ and $\mathbf{W} = (-c, N)$,

$$\det[\mathbf{v}, \mathbf{W}] = \det \begin{pmatrix} p & q \\ -c & N \end{pmatrix} = Np + cq = 13p + 3q. \quad \square$$

The law is anchored to reality by its two fixed points:

$$n(1, 0) = |13| = 13 \quad (\text{the 13 protofilaments}), \quad n(0, 1) = |3| = 3 \quad (\text{the 3–start helix}),$$

which are precisely the two helical families observed by cryo–electron microscopy.

Corollary 1 (No Fibonacci privilege). *Since $\gcd(N, c) = \gcd(13, 3) = 1$, the map $(p, q) \mapsto Np + cq$ is surjective onto $\mathbb{Z}$; every positive integer occurs as a start number for some primitive direction. In particular the Fibonacci values 5 and 8 occur, but so do 2, 4, 6, 7, 9, 10, $\dots$; no start number is arithmetically distinguished within the family structure of L .*

Proof. By Bézout, $13p + 3q$ attains every integer; primitivity of a representative (p, q) can always be arranged, e.g. $5 = 13(-2) + 3(7)$ with $\gcd(-2, 7) = 1$ and $8 = 13(-1) + 3(7)$ with $\gcd(-1, 7) = 1$. $\square$

Remark 1 (Metric prominence). Corollary 1 concerns *which start numbers exist*. *Physical* prominence is a metric matter: a family is a bond of the structure only if its generating vector is short. Under standard parameters ($h = 4$ nm, circumference ≈ 72 nm so $s \approx 5.5$ nm) the shortest generators are $(1, 0)$ with $|\mathbf{v}| = 4.0$ nm (the 13–start protofilaments) and $(0, 1)$ with $|\mathbf{v}| \approx 5.6$ nm (the 3–start helix); the shortest generators of the 5– and 8–start families have length ≈ 33 nm and ≈ 39 nm respectively. Thus the short structural bonds realise start numbers 3 and 13, not 5 and 8.

3 Lemma 2: the structural invariants in $\mathbb{Z}[\omega]$

Let $\omega = e^{2\pi i/3}$, so $\omega^2 + \omega + 1 = 0$ and $\omega^3 = 1$. Write $\mathbb{Z}[\omega]$ for the Eisenstein integers, with norm $N(a + b\omega) = a^2 - ab + b^2$, and let χ_3 denote the nontrivial character of $(\mathbb{Z}/3)^\times$ extended by $\chi_3(3) = 0$; equivalently χ_3 is the Legendre–type symbol governing splitting in $\mathbb{Q}(\sqrt{-3})$: for a rational prime $\ell \neq 3$, $\chi_3(\ell) = +1$ iff ℓ splits ($\ell \equiv 1 \pmod{3}$) and -1 iff ℓ is inert ($\ell \equiv 2 \pmod{3}$).

Lemma 2 (Arithmetic type of the invariants). *The two coefficients of the start–number form of Lemma 1 are the arithmetically distinguished primes of $\mathbb{Q}(\sqrt{-3})$:*

- (i) *The protofilament number $13 \equiv 1 \pmod{3}$ splits: $13 = N(4 + \omega) = (4 + \omega)(4 + \bar{\omega})$, and $\chi_3(13) = +1$.*

- (ii) The start number 3 is the ramified prime: $3 = -\omega^2(1-\omega)^2$ with $N(1-\omega) = 3$, and $\chi_3(3) = 0$.
- (iii) The Fibonacci indices satisfy $5 \equiv 2$ and $8 \equiv 2^3$ with $2 \equiv 2 \pmod{3}$; both are inert, unrepresented by $a^2 - ab + b^2$, with $\chi_3(5) = \chi_3(2) = -1$.

Proof. (i) $4^2 - 4 \cdot 1 + 1^2 = 16 - 4 + 1 = 13$, so $N(4 + \omega) = 13$; as $13 \equiv 1 \pmod{3}$ it splits in $\mathbb{Q}(\sqrt{-3})$ and $\chi_3(13) = +1$.

(ii) Expanding, $(1 - \omega)^2 = 1 - 2\omega + \omega^2 = 1 - 2\omega + (-1 - \omega) = -3\omega$, whence $-\omega^2(1 - \omega)^2 = -\omega^2(-3\omega) = 3\omega^3 = 3$. Thus 3 is a unit times $(1 - \omega)^2$, i.e. $(1 - \omega)$ is the unique ramified prime of $\mathbb{Q}(\sqrt{-3})$ (the field of discriminant -3), with $N(1 - \omega) = 1^2 - 1 \cdot (-1) + (-1)^2 = 3$ and $\chi_3(3) = 0$.

(iii) $5 \equiv 2$ and $2 \equiv 2 \pmod{3}$ are inert, so $\chi_3(5) = \chi_3(2) = -1$; the positive-definite form $a^2 - ab + b^2$ represents $0, 1, 3, 4, 7, 9, 12, 13, \dots$ (namely 3 and the primes $\equiv 1 \pmod{3}$ together with their products) and never represents 2 or 5. $\square$

4 Synthesis

Theorem 1 (χ_3 governs the helical arithmetic). *The start-number law of the 13_3 microtubule lattice is a χ_3 -form: in $n(p, q) = |Np + cq|$ the ordered pair $(N, c) = (13, 3)$ consists of a split prime and the ramified prime of $\mathbb{Q}(\sqrt{-3})$. Moreover the two structurally realised families are the two special arithmetic types: the 13-start protofilament family is the split prime $13 = N(4 + \omega)$, and the 3-start helix is the ramified prime $(1 - \omega)$, whose residue field is $\mathbb{Z}[\omega]/(1 - \omega) \cong \mathbb{F}_3$. The three strands of the 3-start helix form a torsor under the order-3 rotation $\langle \omega \rangle \subset \mathbb{Z}[\omega]^\times$, i.e. the cube roots of unity $\{1, \omega, \omega^2\}$ — the $\{0^\circ, +120^\circ, -120^\circ\}$ “tri-phase.” The Fibonacci indices 5, 8 are inert ($\chi_3 = -1$) and, by Corollary 1 and Remark 1, are neither arithmetically nor metrically distinguished.*

Proof. Immediate from Lemmas 1 and 2: the coefficients $(13, 3)$ receive their arithmetic type from Lemma 2, and the realised families $n(1, 0) = 13$, $n(0, 1) = 3$ are the split and ramified primes. The Eisenstein units are $\{\pm 1, \pm \omega, \pm \omega^2\}$, a cyclic group of order 6; multiplication by ω is the 120° rotation and permutes the three lateral strands cyclically, giving the stated $\mathbb{Z}/3$ action. $\square$

Remark 2 (Consistency with the class-number-one representation identity). Because $\mathbb{Q}(\sqrt{-3})$ has class number one, the number $r(k)$ of representations of k by $a^2 - ab + b^2$ satisfies $r(k)/6 = \sum_{d|k} \chi_3(d)$. Evaluated at the invariants this returns their arithmetic type transparently:

$$\frac{r(13)}{6} = \chi_3(1) + \chi_3(13) = 2 \text{ (split)}, \quad \frac{r(3)}{6} = \chi_3(1) + \chi_3(3) = 1 \text{ (ramified)}, \quad \frac{r(5)}{6} = \chi_3(1) + \chi_3(5) = 0 \text{ (inert)}.$$

Remark 3 (Scope: arithmetic, not metric). Theorem 1 is a statement about the two integer invariants $(N, c) = (13, 3)$ and the $\mathbb{Z}/3$ screw symmetry of the 3-start helix, *not* a claim that the tubulin surface lattice is isometric to the hexagonal Eisenstein lattice. Under the parameters of Remark 1 the interior angle between $\mathbf{a}_1$ and $\mathbf{a}_2$ is $\approx 80.5^\circ$ (oblique, near-rectangular), not 60° ; the Eisenstein structure enters through the *arithmetic* of the start-number form and the cube-root-of-unity symmetry, not through a metric identification.

Remark on “Fibonacci governance.” The recurrent claim that the microtubule is governed by the triple $(5, 8, 13)$ with $5 + 8 = 13$ originates in *phyllotaxis*: a cylindrical point set seeded at the golden divergence angle 137.5° has its two nearest parastichy families indexed by consecutive Fibonacci numbers. The 13_3 microtubule has divergence $2\pi/13 = 27.7^\circ$ and is not a golden-angle lattice; by Corollary 1 its start numbers realise *every* integer, and by Remark 1 its short bonds are

the 3- and 13-start families. The load-bearing invariants are thus the χ_3 -distinguished primes 13 and 3, not the inert Fibonacci pair.

References

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